

Lab 07: Network Protocols

Operating Systems and Distributed Systems

The purpose of this lab is to introduce tools that provide valuable information when working with networks. Additionally, it enforces the students' understanding of the layers in both the OSI and TCP/IP models.

The lab is structured so the student is assisted through the tasks.

Remember to read the exercise and the accompanying text thoroughly before asking for help.

Contents

Exercise 01: Examine the tools	2
Exercise 02: Take your own route	4
Exercise 03: Networking Models	5
Exercise 04: To Send a Package	6
Exercise 05: NAT	7

Exercise 01: Examine the tools

In this exercise, the student is going to use the Linux tools `ip` and `netstat`.

The tools are used to configure network interfaces, manipulate the routing table and show network status.

`$ ip link` is used to show the devices interfaces.

`$ ip address` is used to show the ip's that the device's interfaces hold.

`$ ip route` is used to show the networking routes of the device.

`$ netstat` is used to show current network connections.

Your task

1. Examine the commands `ip link`, `ip address`, `ip route` and `netstat` and document their purposes and how they work.

The exercise will return the following message:

```
1 $ ip link
2 1: wlp0s20f3: <BROADCAST,MULTICAST,UP,LOWER_UP> state UP group default
3     link/ether a3:12:03:ff:6c:c3 brd ff:ff:ff:ff:ff:ff
4 2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> state UP group default
5     link/ether ff:e0:4e:6a:ff:d5 brd ff:ff:ff:ff:ff:ff
6
7
8 $ ip address
9 1: wlp0s20f3: <BROADCAST,MULTICAST,UP,LOWER_UP> state UP group default
10    link/ether a3:12:03:ff:6c:c3 brd ff:ff:ff:ff:ff:ff
11    inet 10.94.100.100/24 brd 10.94.127.255 scope global wlp0s20f3
12    inet6 fe80::7bed:a5b1:82d5:91ae/64 scope link noprefixroute
13 2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> state UP group default
14    link/ether ff:e0:4e:6a:ff:d5 brd ff:ff:ff:ff:ff:ff
15    inet 192.168.10.11/24 scope global enx00e04e6a27d5
16        valid_lft forever preferred_lft forever
17
18
19 $ ip route
20 default via 10.94.100.1 dev wlp0s20f3 proto dhcp metric 600
21 10.94.100.0/19 dev wlp0s20f3 kernel scope link src 10.94.100.100 metric 600
```

```
22 192.168.10.0/24 dev eth0 proto kernel scope link src 192.168.10.11
23
24 $ netstat
25 Active Internet connections (w/o servers)
26 Proto Recv-Q Send-Q Local Address           Foreign Address         State
27 tcp        0      64 192.168.10.11:ssh       192.168.10.10:56600    ESTABLISHED
```

Exercise 02: Take your own route

In this exercise, two or more people are going to connect their computers either directly, through a switch or a mobile hotspot.

Important 1

Disable automatic IP assignment in your network settings.

Hint 1

Assignment of IP addresses to network interfaces can either be done through the GUI or tools like `ip`.

Hint 2

Creation of routes can be done using the `ip` tool.

Your task

1. Connect your computer to another computer, either directly, through a switch or a mobile hotspot
2. Assign an IP address to your network interface
3. If no route is created, then a route entry has to be created to enable communication with the new network
4. Spin up the Docker Compose exercise from Lab 02
5. Let someone connected to your computer (either directly or indirectly) visit your Docker Compose website through your selected IP address

Exercise 03: Networking Models

In this exercise, the student is going to investigate the two network models (OSI and TCP/IP), explain how they work and what happens at each layer.

Your task

1. Discuss the differences between the OSI model and the TCP/IP model
2. Which layers in the OSI model correspond to which layers in the TCP/IP model
3. What is the responsibility of each layer?
4. What does it add or remove?
5. How does addressing work on the layer? Do you know some details about how the layer is implemented?

Exercise 04: To Send a Package

In this exercise, the student is going to explain how packages are sent over a network.

Your task

1. Explain step by step how a package from Computer A with the ip 185.8.135.136 gets to Computer B with the ip 101.24.34.2
2. Which layers in the TCP/IP model does communication happen through?

Exercise 05: NAT

In this exercise, the student is going to explain what Network Address Translation (NAT) is and how it works. You may have to research a little to answer the questions.

Your task

1. Explain what NAT is
2. How does NAT work?
3. When is NAT useful?